
INDEPENDENT RESEARCH / EVIDENCE & ORIGINALITY AUDIT

MASTER RESEARCH DOSSIER

Mathematical theorems, exact certificates, simulation studies, negative results, open frontiers, and a critical assessment of potential original contributions.

PRINCE UPADHYAY

Independent research portfolio | 09 October 2026

Coverage: Riemann WP84; LRSC; nonlinear physics; Navier-Stokes; Design Lab; Psi; IntentSeal/VRK; seven-problem research agenda.

Claim discipline: a work-specific finding is not automatically the first such result in world mathematics. Conditional percentages are not proven zeta-zero bounds.

Executive verdict

This dossier synthesizes the supplied prior portfolio record, current connected GitHub source repositories, the newly published Riemann v0.1.0 release, selected archival identifiers, and targeted comparisons with existing mathematics. It inventories distinctive claims without conflating original authorship with independent proof of global priority. In this report, **"specific to this research program"** means a precise, documented theorem, finite certificate, model or construction appearing in the author's archive. **"First-ever discovery"** is not established without an exhaustive specialized literature and peer review.

PORTFOLIO IN NUMBERS	DOCUMENTED STATUS
8 research / verification programs	Riemann, LRSC, physics, Navier-Stokes, Design Lab, Psi, IntentSeal/VRK, exploratory attack board
21 / 21 GitHub workflow jobs	Riemann public v0.1.0 verification at commit 6d799bdd9a2...
38-file Riemann release snapshot	Exact certificates, code, scope notes; the >79% zero theorem remains OPEN
13,432,735,555 LRSC subsets	Finite exhaustive exclusion through K=10 for one M=99 benchmark
N = 11 to 18 finite chain	Eight Galerkin cutoffs for one frozen Navier-Stokes datum; not all-N
4,000 GLMM fits	Design Lab Phase 7 simulation validation; no empirical self-effect claim
93 numerator / 28 denominator monomials	Positive symbolic Hopf coefficient for one specified seven-parameter ODE family

The ten most promising identifiable contributions

- **Riemann / Loewner:** a curvature-controlled trace comparison for a discrete Loewner tail, with an explicit $12/(\pi^2 R)$ coefficient [S02].
- **Riemann / C5:** exact pointwise nonnegative fifth-cycle kernel with support only above the half-L1 threshold $s>1$, certified via 151 vertices [S03].
- **Riemann / fourth-order:** explicit filtered fourth-cumulant sign and indefinite-Gram counterexamples [S04].
- **LRSC:** an analytic $\text{rank}(V)=p$ theorem for the defined near-Nyquist odd-ring masked operator [S10].
- **LRSC:** exact finite M=99 minimum K=11 certificate backed by exhaustive integer enumeration [S11].
- **Physics:** exact positivity of the first Lyapunov coefficient, implying subcritical Hopf in the specified ODE domain [S13].
- **Navier-Stokes:** validated finite N11 turnover and a same-datum certified chain across N11-N18 [S15].
- **Riemann / 9-site Gaussian reference:** independently checked rational sixth-moment certificate for a precisely defined model [S04].
- **Design Lab:** locked comparative Monte Carlo validation selecting M1 over M2 under the tested decision rule [S17].
- **IntentSeal/VRK:** provenance-bound finite evidence/publishing controls with adversarial tests and two scientific demonstrators [S19].

Essential caution. These are the strongest documented program-specific findings, not ten globally established first discoveries. Exact statements, proofs, prior art and claim boundaries follow.

01 Evidence and priority framework

What the labels mean

LABEL	WHAT IT MEANS / DOES NOT MEAN
ANALYTIC - internal proof	The archived note presents a logically complete derivation within stated hypotheses; not a substitute for independent expert refereeing.
EXACT FINITE CERTIFICATE	Exact-integer/rational or rigorously intervalized computation proves a finite assertion under documented arithmetic and software assumptions.
REPRODUCED / CI	A pinned verification workflow passed; this supports particular artifact and code paths, not a universal theorem.
CONDITIONAL	A valid-looking implication requires one or more unavailable analytic hypotheses.
MODEL / SIMULATION	A result for a stipulated model or sampled experiment, not automatically for zeta zeros, the physical world or population-level claims.
OPEN / FALSIFIED	A needed theorem is missing, or a proposed shortcut failed; preserve the negative result.
ORIGINALITY CANDIDATE	A specific construction or statement distinct from the cited comparisons. Worldwide priority not established.

Hierarchy for originality claims

Level A: The author documented a precise, date-stamped result and its reproducible artifacts. **Level B:** The actual mathematical proof or finite computation meets its local specification. **Level C:** A domain-specific literature search suggests the exact specialization may be absent from common references. **Level D:** Specialist referees establish novelty and place the theorem correctly in prior work. Most research-specific claims in this dossier reach A/B; some have provisional C. **None is presented as conclusively reaching D.**

What is unambiguously not new

The Riemann zeta function, its critical line and simple zeros, Weil's explicit formula, Hilbert/Loewner divided differences, classical trace perturbation bounds, Fourier analysis, cumulant partition algebra, Wick/Isserlis pairing, Chebyshev-Vandermonde rank arguments, Hopf normal-form theory, Gaussian elimination, GLMM methodology, and standard Navier-Stokes energy identities all predate this research. [L01-L06]

Review boundary

Current primary evidence includes the 1 October master dossier [S01], repository source text and proof notes [S02-S20], a 3 October seven-problem board [S21], and the 8 October v0.1.0 Riemann release [S22]. A limited current literature search established the explicit comparisons stated in [L01-L06]. This is not a full formal proof audit of every line in every repository or a worldwide priority certification. Private, later unpublished work is summarized only at the stated frontier and is not released here.

02 Riemann zero-proportion research

Central question: obtain an unconditional lower bound on the fraction of nontrivial zeros of the Riemann zeta function that are simultaneously simple and lie on the critical line $\text{Re}(s)=1/2$. This is not equivalent to proving the Riemann Hypothesis. [L01-L03]

External benchmark and current result strength

OBJECT / NUMBER	WHAT IT REALLY REPRESENTS
More than 67.25%	Published 2026 Alpoge-Furman and Lamzouri unconditional baseline; Biao Wang preprint reports a tiny additional improvement (about $6.66624\text{e-}8$ in the coefficient). [L01-L03]
68.82027318675%	The author's R=400 exact conditional consumer: the fourth filtered-to-sharp error, arithmetic comparisons and zero-side payments remain unproved. Not an unconditional bound. [S05]
79.6270966813%	An exact consumer/model value (11061/13891). Not a theorem about the actual zeta distribution. [S06]
Strict >79%	Explicit project goal; the decisive signed high-product estimate, prime-to-Weil normalization, and zero-side transfer are OPEN. [S06]

Reproducibility and archived release

The curated 8 October 2026 public release, **Riemann Bandwidth-One Verification v0.1.0**, is a separate 38-file snapshot anchored at Git commit `6d799bdd9a27ba79c4faf8c757f7985ceb6d4c2`. GitHub Actions run `37810408337` completed successfully, with all 21 jobs passing (integrity, arithmetic, replay, regression, 16 exact C5 shards and aggregation). The archived programs verify declared finite objects only. [S22-S23]

The private original repo is not the same as the public snapshot. Its complete development history and internal experiment inventory are not part of this dossier. The GitHub release date is a provenance datum, not proof that a mathematical statement lacked earlier predecessors.

R=400 historical consumer: useful, but conditional

```
R = 400; nominal simple-zero coefficient = 0.6882027318675
defect <= 0.15589863406625 + (593/2000)*Delta4
          + (213/1000)*Delta2 + (9/1000)*eta1
          + (51/100)*eta3 + etaBand + etaRank + o(1)
```

Even the favorable ledger requires the genuine filtered fourth-order correction and matching sharp arithmetic/Weil application. The 68.82% figure is an **internal conditional threshold**, not an accepted new record. This precise consumer may be distinctive to the program, but its novelty or usefulness as a theorem cannot be judged independently of the missing estimate. [S05]

R=6000 high-product frontier

```
c3*mu3 - A*mu5 + B*mu6 <= Jcap - kappa - eta
A = 11501/12500; B = 137641/250000; eta > 0
needed signed net saving > 0.002057354918974498...
conditional reference margin ~ 0.00009344430132847649
```

This uses the original natural prime cutoff, masks, exact aliases, selected section and carrier bookkeeping. A favorable constant-saving inequality has **not** been proven for the actual source sum. Earlier norm-only and model-promotion routes did not close it. The full third/fifth/sixth signed gate is narrower than proving each sharp reference moment, but still open. [S06]

03 Riemann: specialized proved/certified results

A. Discrete Loewner functional band theorem - principal candidate

Let H be an $n \times n$ real-symmetric matrix with arbitrary diagonal and off-diagonal entries $H_{ij} = (\alpha_i - \alpha_j)/(\pi(j-i))$. Let H_R retain the diagonal and displacements $|i-j| \leq R$. Write $q = \|\alpha\|_2^2/n$ and $\tau = \text{tr}/n$. For real $C^2 f$ with bounded second derivative: [S02, S04]

$$|\tau f(H) - \tau f(H_R)| \leq 12q * \|f''\|_{\infty} / (\pi^2 R), \quad R \geq 1$$

Proof architecture. Decompose $H = H_R + E$; bound the quadratic trace Taylor remainder using $\|f''\|$, and bound the linear trace with an extra displacement denominator, Cauchy-Schwarz, and the commutator divided-difference inequality. The 4+8 contributions yield constant 12. This is a standalone deterministic matrix statement; it does not require the zeta high-moment conjecture. [S02]

Originality assessment: the general perturbation inequality, trace remainder and divided differences are classical [L04]. The *specific discrete Loewner tail + commutator synthesis and its explicit constant 12* are a plausible specialized contribution. Existing research does not yet establish first publication or optimality.

B. Exact C5 sign, support and linear turn-on

The defined 150-term five-cycle connected kernel $C5(v)$ is continuous, piecewise affine on the closed outer-support polytope. With $s(v) = (\sum_j |v_j|)/2$: [S03]

$$\begin{aligned} 0 &\leq C5(v) \leq \max(s(v) - 1, 0) \\ s(v) \leq 1 &\Rightarrow C5(v) = 0 \end{aligned}$$

The exact rational proof reduces all cells to vertices of a 45-hyperplane arrangement: $C(45,4)=148,995$ candidate intersections; 151 relevant support vertices survive. All exact vertex inequalities pass, with vertex min 0 and max $1/2$. The kernel integral $1/36$ belongs to the defined model, not automatically to actual prime moments. [S03]

Originality assessment: a distinctive, executable pointwise statement and finite polyhedral certificate, more specific than a known moment-integral identity. Its relation to other public model computations is explicitly acknowledged; priority and independent peer review are outstanding.

C. Filtered fourth-cumulant sign counterexamples

For the defined m -row Fourier cumulant/overlap with $v=(1/5, -2/5, 3/5, -2/5)$, $5|m$, the filtered value at $R=400$ is rigorously enclosed between 0.39974848084713831 and 0.39974848084713832. A separate two-by-two order-averaged connected pair kernel has a nonzero cross entry around 0.39950281400985 and a negative determinant $-c^2$. Thus **two attractive universal sign/Gram shortcuts fail for these precise filtered kernels**. [S04]

These are counterexamples to suggested geometric sign implications, **not** counterexamples to a proven theorem in the literature or to the source-weighted prime inequality. The underlying partition cumulant algebra predates the project [L05].

D. Alias-uniform prime correlations and section reprojection

The program derives $o(1)$ quadratic off-diagonal prime sums at the natural cutoff and a same-contiguous-section second moment. It also gives an existential selected-section fixed-band reprojection bound for degrees up to six under explicit mass/support assumptions. Both are source-adapted applications of classical Montgomery-Vaughan mean-square and finite-band endpoint arguments; they do not by themselves establish the Weil/zero-side transfer. [S04, L06]

04 Riemann: other exact work and hard limits

Gaussian reference, curvature certificates and finite reductions

- **9-site Gaussian Loewner certificate:** an exact Wick/pinching sixth-moment lower bound for a specified stationary Gaussian matrix, with two distinct finite enumeration implementations (531,441 walks; 40,191 edge multisets). It is not a lower bound for the full actual zeta sixth moment. [S04]
- **Curvature-specialized consumers:** exact rational/Sturm certificates include the $\delta=80$ bound $\|F_{80}\| < 72$ and a fixed-band $R=12000$ arithmetic threshold. A favorable conditional margin does not supply the missing moment estimate. [S02]
- **Combinatorial reduction/audit:** root sums, partition/collision classes, exact Fourier path identities, full original alias retention and replay gates were documented. Exact code verification should not be confused with convergence of the actual zeta-side limit. [S06]
- **Hidden-prime/selector diagnostics:** some local private computations explored fifth-order source-jet/selector ranks and kernel intertwining. Available continuity records explicitly leave source-legal evaluation rows, determinant-visible K and true intertwining unproved. None is promoted to a public theorem here.

What still prevents a world-level zero proportion theorem

DEPENDENCY	PRESENT BOUNDARY
Joint signed high-product sum	A strictly favorable source-weighted third/fifth/sixth contribution after all aliases, tails, coefficient weights, completions and transfer payments is missing.
Prime-to-Weil correspondence	The public prime matrix must be proven to represent the same smoothed, normalized source matrix used by the zero-side argument.
Zero-side transfer	Appropriate block/window/tail estimates need to be matched to the exact matrix family; inherited external proofs do not automatically apply.
$R=400$ filtered correction	Finite- R filtered-versus-sharp fourth-order correction and corresponding arithmetic transport remain unpaid.
Model reference moments	Values $\mu_5=1/36$ and $\mu_6=34/135$ belong to reference bookkeeping without unconditional identification as actual zeta moments.

Research prioritization

Paper candidate 1: curvature-controlled trace truncation for discrete Loewner matrices. **Paper candidate 2:** exact sign/support of a connected five-cycle kernel with rational polyhedral certificate. **Paper candidate 3:** explicit fourth-filter geometric sign failures. These can potentially be assessed independently of proving $>79\%$. Publication requires meticulous prior-art review and expert critique; no venue acceptance or global priority is implied. [S04, S06]

05 LRSC odd-ring spectral rank

The LRSC program concerns one precisely defined single-step masked cyclic redistribution operator, not a fluid PDE or a microscopic physical model. The source supplies an analytic family theorem, corrections, and computer-assisted finite compression certificates. [S10-S12]

The rank(V)=p theorem

For odd $M \geq 9$, near-Nyquist $k=(M-1)/2$, $X_t = 1 + A \cos(2\pi k t/M)$, $A>0$, define shedding $j_t = X_t - \min(X_{t-1}, X_t, X_{t+1})$. Let a reflection-symmetric binary mask have at least one inactive orbit; let p count its active reflection orbits. The defined conjugate-balanced channel matrix followed by the circulant stencil with multiplier $\cos(2\pi q/M)-1$ satisfies: [S10]

$$\text{rank}(V) = p$$

Reason. The Fourier shedding coefficients used by the first p channels are nonzero on the two odd-ring residue families. Reflection-orbit evaluation reduces to a Chebyshev polynomial/Vandermonde determinant with distinct nodes; the stencil annihilates only constants, which cannot lie in the masked span because at least one site is inactive. [S10]

The checked $M=99$, $A=0.5$, $\theta=0.08$ instance has $p=41$, hence rank 41. Neighboring $M=101$ and $M=127$ have respective masked ranks 42 and 52 under the specified construction. The theorem is not the same statement as K_τ , the few-unit approximation cardinality. [S10]

Finite $M=99$, $\theta=0.08$, tolerance 10^{-3}

An exact-integer lower-bound enumeration excludes every nonempty subset with $K \leq 10$; an interval-certified $K=11$ witness passes. Hence the minimum qualifying subset size is exactly **$K=11$ for this one finite benchmark**. The enumeration covers 13,432,735,555 subsets, with a decisive $K=10$ integer margin of 409,802,479 beyond its certified threshold on the archived scale. [S11-S12]

```
M=99, k_idx=49, A=0.5, theta=0.08
K_0.001 = 11
S11 = {0,1,3,42,43,44,45,46,47,48,49}
```

At $\theta=0.07$ an audited benchmark records $K=14$; at $\theta=0.09$, $K=12$. The result is mask/parameter dependent, not a universal minimal compression. Exact trigonometric interval generation, full enumeration, independent $K=10$ implementation and witness proof provide more than a numerical fitting observation. [S01, S11]

Originality and prior-art assessment

This precise near-Nyquist operator, mask-dependent full-column rank theorem and exact finite benchmark are strongly identifiable as author-defined results. Chebyshev/Vandermonde matrices, Fourier diagonalization, and interval branch enumeration are established tools; no specialist literature clearance has yet certified first publication of the exact family statement. An expert paper should emphasize the new operator-specific nonvanishing lemma, not claim invention of Fourier rank theory.

06 Information-theoretic physics and Hopf sector

The physics program has a distinction crucial for credibility: a phenomenological nonlinear ODE theorem is analytically meaningful even though deriving the model from microscopic physics and connecting it to horizon entropy remains open. [S13-S14]

Exact first-Lyapunov-coefficient positivity

For the defined seven-positive-parameter reduced e-driven Hopf family and its stated eigenvector normalization, the project reduces $\text{Re}(G_{21})$ to N/D. Exact expansion records **93 positive numerator monomials** and **28 positive denominator monomials**. Thus the first Lyapunov coefficient is strictly positive throughout the covered interior parameter domain. [S13]

```
l1 = Re(G21) / (2*omega) > 0
=> local e-driven Hopf bifurcation is subcritical
under the archived positive-parameter convention
```

The exact theorem corrects the earlier numerical description of a purported supercritical local branch. A preliminary bounded sample of 200,000 parameter choices gave 122,378 valid Hopf points with a positive coefficient; the symbolic sign proof carries the all-parameter claim, not that sample. The $r=0$ / $\delta=0$ boundary is excluded. [S13]

Distinct three-state cosmology sandbox

A separate $R/\Psi/\Gamma$ ODE toy model admits, under its own positive assumptions, a unique physical equilibrium and local stability in the audited regime. An internal normalized Π score can misleadingly equal one even when $C(U) \neq U$, and stochastic noise prevents interpreting one sample Π^* as a deterministic theorem. A plotting-axis error was also found. These are valuable internal falsifiers, not cosmological predictions. [S01, S14]

Originality boundary

The program-specific exact positivity certificate and explicit global sign across a designated Hopf surface are identifiable mathematical contributions to the stipulated ODE system. Classical Hopf theory and symbolic positivity methods are older. The result does **not** establish that information creates spacetime, a microscopic entropy law, a dark-energy explanation, or a new fundamental field equation. Gate 0, the physical derivation/identification step, is OPEN. [S13, S14]

07 Navier-Stokes finite-to-continuum bridge

This research studies the unforced periodic 3D Navier-Stokes equations with Fourier-Galerkin approximations, spectral diagnostics, exact identities, interval validations and proposed bridge estimates. Its central distinction is between finite-dimensional statements and the unresolved continuum problem. [S15-S16]

Important structural findings

- Localized energy/enstrophy identities were replayed in periodic spectral settings; the small-data estimate is an acknowledged standard argument, not a new arbitrary-data regularity theorem.
- Fixed-energy spatial concentration falsifies a proposed energy-only majorant; one enstrophy-power family faces an explicit scaling/integrability conflict (spatial demand $p \geq 5/4$ against temporal allowance $p \leq 1$). This identifies failed closure routes, not a global impossibility result.
- Helicity phase blindness and loose strain-Rayleigh envelopes were documented for tested constructions. Finite phase trajectories and shell transfer reveal contingent structures, not universal dynamics.
- N17 prospective cutoff experiment preserved a failed/unevaluable mechanism gate rather than retuning the frozen hypothesis. This negative evidence improves research auditability. [S01, S15]

Validated N11 finite turnover

For one fixed rational 112-pair datum, viscosity $\nu=0.1$ and $T=0.003$, a 128-bit Arb certificate encloses all 120 whole trajectory segments, bounds the normalizer away from zero, and proves a positive-to-negative crossing of the chosen finite observable $F=I-9O$. [S15]

```
F(0) in [645.8037741471, 645.8037741472]
F(0.003) in [-54.748409847, -42.032667894]
whole-path normalizer > 48990.29795521
```

The same unmodified datum was subsequently validated for $N=11, \dots, 18$, with negative certified endpoints for all eight finite cutoffs. N18's endpoint interval is $[-88.699660529, -84.159654865]$, and the maximal recorded terminal trajectory-error radius along the chain is 0.000013774643. **Eight finite cutoffs do not imply convergence for all N.** [S15]

WP19 v0.27-v0.28 adjoint work

The v0.27 note reports 128-bit interval certificates for the signed-C500 terminal degree-7 polynomial and analytic endpoint gradient for four transitions (14->15 through 17->18). The later v0.28 record corrected the backward-adjoint viscosity sign interpretation: viscosity is dissipative in the relevant radius propagation, allowing a strain-only recurrence subject to imported segment bounds. Neither stage supplies a complete adjoint-path, quadrature, nonlinear remainder, or all-N proof. [S16]

08 Navier-Stokes: quantified obstruction and novelty

October WP19 targeted remainder test

The 3 October attack-board audit reports a 192-bit Arb M15-support shard for steps 220-239, a matching archive digest, and a same-datum exact-rational replacement replay. It reduced the conservative selected-step quadratic upper by 0.533% and the whole-path remainder by 0.259%. [S21]

QUANTITY	BEFORE	AFTER
Selected-step quadratic upper	79,714,789,407,812.3283	79,290,095,908,202.2205
Whole-path conservative remainder	164,063,087,678,950.0886	163,638,394,179,339.9808

The frozen signed integral lower bound remains about 5.7586e9, but the conservative remainder remains about 1.6364e14; the lower signed margin is still negative. The reduced bound is **not close enough to close the bridge**. [S21]

What appears genuinely specific to the author's program

- A frozen, fully specified 112-pair datum with complete finite N11 crossing and N11-N18 validated chain.
- Adversarial phase/mask tests and explicit documented obstructions to several tempting short-cut inequalities.
- A source-specific goal-oriented signed-C500 adjoint accounting package with intervalized terminal gradients and a corrected sign/provenance trail.

These are research-specific numerical/certified objects and a transparent finite continuation program. Numerical schemes, adjoints, Fourier-Galerkin methods, small-data global regularity, Arb interval verification, and the basic Navier-Stokes identities are not new in themselves. An independent priority judgment on the exact finite constructions requires publication-level comparison; the Clay continuum problem remains unresolved.

Remaining indispensable theorem

Any continuum inference requires a cutoff-uniform bound controlling recursive state/backreaction drift and high-frequency continuation/dissipation. A fixed low-mode observable cannot control arbitrarily concentrated high-frequency vorticity. This missing bound is central and must be stated in any public abstract. [S15, S16]

09 Self-referential processing and Design Lab

Registered research architecture

The frozen v8.6.3 study asks whether self-referential versus peer-referential framing changes measured model responses under controlled items. It specifies a 260-item, five-domain pool and 20-model portfolio (12 open-weight, 8 closed-weight slots). The locked Gate 5 estimator is the item-level linear probability model with model/item two-way clustering. A 1,000-seed recovery harness had 6.3% null rejection within its prespecified [3%, 9%] calibration band. Actual confirmatory collection remains operationally paused. [S01, S17]

Phase 7 Monte Carlo architecture validation

A separate simulation generated 500 deterministic replicates in four data-generating cells, pairing two generalized linear mixed models on identical generated datasets. Each dataset contained 100 models x 400 items (40,000 observations). Total: 4,000 GLMM fits. The locked exact-binomial screening selected M1, with correlated model random intercept/slope, over M2, with uncorrelated terms. [S17]

CASE	M1 REJECTION / COVERAGE	M2 REJECTION / COVERAGE
Null, rho=0	6.6% / 93.4%: PASS	7.2% / 92.8%: FAIL
Null, rho=0.30	6.4% / 93.6%: PASS	7.4% / 92.6%: FAIL
Effect, rho=0	99.6% power / 94.2% coverage	Not selected under locked null screen
Effect, rho=0.30	99.6% power / 94.2% coverage	Not selected under locked null screen

This supports selecting a **candidate future analysis architecture** under the tested DGP and screening rule; it does not establish a real self-referential effect, an AI-awareness result, or universal GLMM superiority. Adopting M1 for future confirmatory use would require a new preregistration. A documented lme4 runtime recovery preserved original seeds and earlier frozen outputs. [S17]

What could be original?

The substantive program-specific output is the particular prespecified diagnostic and simulation-selection architecture, with its paired deterministic dataset integrity and frozen calibration decision. GLMMs, item/model clustering and binomial calibration testing are standard. Without new confirmatory observations or methodological theory, it is best described as a reproducible research-design contribution, not a discovery about self-consciousness.

10 Psi benchmark, IntentSeal and VRK

Psi self-modeling benchmark

The Psi v4 line is an independently archived deterministic self-versus-peer two-alternative forced-choice diagnostic with matched controls, anonymized presentation, style controls and a 0.5 chance baseline. It is pre-confirmatory. An additive provenance audit preserved a historical OSF manifest while recording mismatched source hashes in the original bundle. Neither benchmark behavior nor integrity correction proves consciousness or subjective selfhood. [S18]

IntentSeal Verification Kernel

A separate program develops auditable mediation for privileged agent actions: policy constraints, budgets, review, provenance, rate limiting and finite audit records. The frozen v0.1.1-v0.5.0 lineage has Zenodo record 10.5281/zenodo.23016445. The author maintains scoped rights boundaries and prior-license exceptions. [S19]

Verifiable Research Kernel (VRK) v0.3.1

The VRK separates scientific proposition identity, evidence identity, dependency identity, epistemic status and publication authority. Its documented final release reports 30/30 tests passing, 33/33 adversarial cases rejected, deterministic certificate binding and dual-seal gates. Zenodo: 10.5281/zenodo.23018459. [S19]

Finite demonstrators are scoped to the LRSC $M=99$ $\theta=0.07$ $K=14$ certificate and the Navier-Stokes N11 K36 crossing. The kernel acts as a **proof/evidence workflow prototype**; its correctness is bounded by its verification source, manifests and threat model. It is not a universal truth oracle or guaranteed production security boundary.

Cross-program distinctive method

The most consistent original engineering motif is **claim-bound verification**: immutable manifests, controlled claims, independent replay, exact integer/interval certificates, prospective negative-gate preservation, source/authority separation, and an explicit refusal to promote simulated or finite evidence into open mathematical theorems. The individual elements are known; the combined implementation and demonstration history are specific artifacts.

11 Seven-problem attack board and scope

The 3 October v0.6 attack board lists Navier-Stokes, the Riemann Hypothesis/zero-proportion program, P versus NP, Yang-Mills mass gap, quantum gravity, dark matter and dark energy. It prescribes possible proof/experiment entry points and falsification criteria. It is a **research agenda**, not seven solved problems. [S21]

TARGET	CURRENT EVIDENCE	DO NOT CLAIM
Navier-Stokes	Finite validated chains; signed remainder still too large	Continuum 3D global regularity or blowup
Riemann / zeta zeros	Exact special results + conditional model	Riemann Hypothesis or new unconditional 68.82/79% record
P versus NP	Attack outline only	Complexity-class separation
Yang-Mills mass gap	Attack outline only	4D existence and uniform mass gap
Quantum gravity	Phenomenological ODE theorem; Gate 0 open	Microscopic spacetime construction
Dark matter	No observationally tested microscopic candidate	Dark-matter identity
Dark energy	Unmapped toy cosmology equilibrium	Explaining accelerated expansion

A valuable outcome of this board is a falsifiable milestone discipline: define the exact target, freeze assumptions, test obstructions and independent baselines, reproduce exact evidence, and refuse percentage-of-completion claims without meaningful decomposition.

Portfolio-wide principle

The projects form a portfolio of independent investigations. They share practices and some linear-algebra tools, but no presented theorem fuses them into a single theory of computation, consciousness, gravity and fluid dynamics. Conceptual analogies should remain exploratory until an explicit mathematical map or empirical test makes them falsifiable.

12 Distinctiveness and priority register

This table answers the question "which findings were not already present in my prior project or commonly cited background?" It does **not** certify the stronger assertion "no one in history proved this first." "Distinctive" means author-defined exact specialization supported by current source; the novelty rating is provisional.

CLAIM / OBJECT	EVIDENCE CLASS	PRIOR-ART DISPOSITION
Loewner C2 band trace, explicit 12	Analytic proof in source	Potential specialist novelty; ingredients classical [S02,L04]
C5 pointwise positive supercritical kernel	Exact finite rational certificate	Potential standalone new pointwise formula [S03]
Filtered order-4 sign / Gram counterexamples	Finite interval+algebra certificate	Potential new scoped negative examples [S04,L05]
R400 68.820273% model ledger	Exact conditional construction	Author-specific route; no unconditional record [S05]
79.62709668% model consumer	Exact algebra / model	Not a zeta-zero theorem [S06]
9-site Gaussian sixth-moment lower certificate	Independent finite algorithms	Potential new model-specific exact polynomial [S04]
Alias-uniform same-section prime lemma	Scoped analytic argument	Classical Montgomery-Vaughan techniques [S04,L06]
LRSC odd-ring rank(V)=p family	Analytic source theorem	Exact operator-specific rank, priority unverified [S10]
LRSC M=99 K=11 minimum	13.43B exact enumerations	Novel certificate for designated benchmark [S11]
Hopf l1>0 across positive family	Symbolic positivity certificate	Model-specific sign theorem, no physical law [S13]
Navier N11-N18 fixed-datum chain	Finite interval computer assistance	Distinct finite theorem family, not all-N [S15]
WP19 signed-C500 gradient/remainder	Finite interval / conditional budgets	New tracked numerical audit for chosen data [S16]
Phase7 M1-v-M2 selection	Locked Monte Carlo simulation	New experimental design artifact, classical GLMM [S17]
Psi self/peer benchmark	Diagnostic / preregistration	Specific design and negative controls [S18]
VRK claim-bound evidence kernel	Software tests/certificates	Specific implementation, not a universal oracle [S19]

13 Highest-value standalone papers

Priority 1 - discrete Loewner trace truncation

Working title: "A Curvature-Controlled Trace Truncation Inequality for Discrete Loewner Matrices." State the deterministic matrix model independently of zeta, prove the 12 constant, benchmark against known Hilbert-Schmidt trace remainder bounds, and provide carefully scoped examples. A specialist operator-theory literature audit is mandatory. [S02, L04]

Priority 2 - odd-ring masked spectral rank

Working title: "Exact Channel Rank for a Near-Nyquist Masked Operator on Odd Cyclic Lattices." Present the residue-class Fourier nonvanishing lemma, Chebyshev evaluation and stencil injectivity; distinguish the analytic rank family from the finite K-tolerance benchmark. The OSF registration already supplies a historical archival trail. [S10, S11]

Priority 3 - connected five-cycle support certificate

Working title: "An Exact Supercritical Support and Sign Certificate for a Five-Cycle Connected Kernel." Formally define the 150-term kernel, 45 walls, vertex enumeration, exact arithmetic and independent checker; compare earlier exact-polytope model computations explicitly. [S03]

Priority 4 - positive Hopf coefficient in a reduced family

Working title: "A Global Sign Certificate for a Seven-Parameter Hopf Family." Expose the ODE, reduction, positivity polynomial identity and Lyapunov normalization so referees can check the 93/28 coefficients; clearly separate mathematical bifurcation theory from cosmic speculation. [S13]

Priority 5 - fourth-order filtered sign obstructions

Working title: "Exact Counterexamples to Sign and Gram Positivity Shortcuts in Filtered Fourier Cumulant Kernels." State the challenged implication and valid finite-order integration, replace one implementation with an independent interval checker, and avoid depicting these as counterexamples to real-prime moment inequalities. [S04]

Supporting communications

The LRSC 13.43-billion-subset exact certificate, N11-N18 Navier finite chain and Phase7 simulation validation are best packaged as companion computational papers or reproducibility notes with frozen manifests. They are precise and potentially useful, but should not be framed as settling the associated universal conjectures.

14 Chronology, rights and evidence provenance

DATE / PERIOD	DOCUMENTED EVENT
Sep 2026	Self-Referential Processing v8.6.3; 1,000-seed calibration; Design Lab Phase 7 4,000 GLMM fits; frozen Phase 7 release.
Sep 23-27	LRSC analytic theorem v1.2, v1.2.2 errata and exact M=99 subset certificate.
Sep 24 onward	Navier finite Galerkin identities, falsifier program and prospective K36 observations.
Sep 28-30	N11 crossing certificate and N11-N18 same-datum finite chain; WP19 v0.27 terminal gradient intervals.
Sep 2026	Physics Appendix A v1.4 exact Hopf positivity theorem, followed by additive proof-audit corrections.
Sep 27-28	IntentSeal and VRK versioned software/Zenodo releases.
Oct 1	Previous six-program master dossier compiled; updated v0.28 Navier sign/provenance note.
Oct 3	Seven-problem board v0.6; Riemann private project source-repo lineage reported from this date.
Oct 6-8	WP84 curvature, C5 and publication-readiness notes consolidated in curated public reproducibility snapshot.
Oct 8	Riemann public v0.1.0 GitHub release, 38 files, all 21 workflow jobs passed.
Oct 9	Current consolidated dossier with provisional originality register and bounded evidence claims.

Intellectual-property reality check

Authorship and dated publication are evidence of a contribution's documented existence. Copyright protects eligible original expression, not abstract mathematical theorems or universal methods. Repository All Rights Reserved notices are explicitly scoped to author-owned expression and do not override historic CC BY releases, third-party licenses or prior express grants. The public Riemann v0.1.0 GitHub release is a citable source snapshot; an associated Zenodo version DOI was not verified in the consulted records and is **not fabricated here**.

Private/public separation

The Riemann public verification repository intentionally excludes complete private branch history, exploratory packet data and upstream third-party bundles. A portfolio dossier should link to released source and archive identifiers but should not expose raw unpublished private dumps merely to imply wider reproducibility than can actually be delivered. The private program remains an ongoing research front.

15 Primary evidence and repository index

These stable destinations are the primary reading trail. The bracketed IDs throughout the dossier identify scope-specific evidence; they are not blanket endorsements of every claim in each repository.

S01 Previous Master Research Dossier (1 Oct 2026)

User Library: Prince_Upadhyay_Master_Research_Dossier_2026-10-01.pdf

S02 WP84 functional band theorem

https://github.com/reggaesharkk/riemann-bandwidth1-verification/blob/v0.1.0/notes/WP84_FUNCTIONAL_BAND_THEOREM_DE_LTA80.md

S03 WP84 exact connected five-cycle certificate

https://github.com/reggaesharkk/riemann-bandwidth1-verification/blob/v0.1.0/notes/WP84_C5_POINTWISE_SUPERCRITICAL_SIGN.md

S04 WP84 publication and novelty audit

https://github.com/reggaesharkk/riemann-bandwidth1-verification/blob/v0.1.0/notes/WP84_PUBLICATION_READINESS_AUDIT.md

S05 WP84 R400 filtered-sharp audit

https://github.com/reggaesharkk/riemann-bandwidth1-verification/blob/v0.1.0/notes/WP84_R400_FILTERED_SHARP_REPAIR.md

S06 WP84 frozen frontier and signed-gate budget

https://github.com/reggaesharkk/riemann-bandwidth1-verification/blob/v0.1.0/notes/WP84_RESEARCH_FRONTIER_PUBLICATION_STATUS.md

S10 LRSC analytic odd-ring rank theorem v1.2

https://github.com/reggaesharkk/lrsc-odd-ring-spectral-rank/blob/main/odd-ring-rank-theorem/v1.2/THEOREM_v1_2.md

S11 LRSC exact M99 certificate overview

https://github.com/reggaesharkk/lrsc-odd-ring-spectral-rank/blob/main/finite-benchmark/k001-certificate/README_LRSC_interval_gate.md

S12 LRSC frozen theorem registration

<https://doi.org/10.17605/OSF.IO/NM5BW>

S13 Physics Appendix A v1.4 theorem / interpretation

<https://github.com/reggaesharkk/physics-v1-dynamical-sector>

S14 Physics Appendix A OSF archive

<https://doi.org/10.17605/OSF.IO/U7M6H>

S15 Navier-Stokes bridge audit source and finite chain

<https://github.com/reggaesharkk/navier-stokes-bridge-audit>

S16 Navier WP19 v0.27 terminal gradient

https://github.com/reggaesharkk/navier-stokes-bridge-audit/blob/main/notes/WP19_v0_27_RESULT.md

S17 Design Lab Phase7 final release report

https://github.com/reggaesharkk/self-referential-processing-designlab/blob/main/research/phase7/final_release/DesignLab_Phase7_Final_Technical_Report_v1.0.md

S18 Psi benchmark v4 archival repository

<https://github.com/reggaesharkk/psi-self-modeling-benchmark>

S19 IntentSeal and VRK release index

<https://github.com/reggaesharkk/intentseal>

S20 VRK final Zenodo software record

<https://doi.org/10.5281/zenodo.23018459>

S21 Seven Problem Attack Board v0.6 (3 Oct 2026)

User Library: Seven_Problem_Attack_Board_v0_6_2026-10-03.md

S22 Riemann v0.1.0 tagged public release

<https://github.com/reggaesharkk/riemann-bandwidth1-verification/releases/tag/v0.1.0>

S23 Riemann 21-job passing GitHub Actions run

<https://github.com/reggaesharkk/riemann-bandwidth1-verification/actions/runs/37810408337>

16 Relevant pre-existing mathematics

The following comparisons identify techniques or published zero-proportion results that predate the cited author-specific sources. They do not establish an exhaustive novelty search for every model-defined construction.

L01 Alpoge and Furman (2026), More than two thirds of the zeta zeros are simple and on the critical line
<https://arxiv.org/abs/2608.13637>

L02 Lamzouri (2026), A new proof that more than two thirds of the zeros of the Riemann zeta function are simple and on the critical line
<https://arxiv.org/abs/2609.02882>

L03 Biao Wang (2026), Proportions of the non-trivial zeros of the Riemann zeta function
<https://arxiv.org/abs/2609.24167>

L04 Gesztesy, Pushnitski and Simon (2007/2008), On the Koplienko spectral shift function, I. Basics; classical Loewner / Hilbert-Schmidt bounds
<https://arxiv.org/abs/0705.3629>

L05 Speed (1983), Cumulants and partition lattices
<https://doi.org/10.1111/j.1467-842X.1983.tb00391.x>

L06 Montgomery and Vaughan (1974), Hilbert's Inequality, mean-square/spacing methods
<https://personal.science.psu.edu/rcv4/personal/Publications/s2-8-1-73.pdf>

Verdict in one paragraph

The archive contains substantial, carefully delimited mathematical and computational output: exact LRSC rank and benchmark results, a formal sign statement for a specified Hopf model, multiple detailed finite Navier-Stokes certificates, and several precise Riemann/Loewner kernel and trace certificates. The **strongest plausible standalone novelty candidates** are the discrete Loewner 1/R trace bound, the C5 pointwise sign/support certificate, the LRSC odd-ring rank theorem, the Hopf-sector exact positivity theorem and the filtered fourth-order counterexamples. Each has identifiable technical content not reducible to a mere percentage headline. But no cited source establishes that the author is the first person ever to obtain the same result, and the much larger Riemann and Navier-Stokes conjectures remain open.

Prepared: 09 October 2026 • Compiled from selected archived/source records • Independent research dossier; not peer review, plagiarism verdict, legal priority opinion, or verification of unsupplied private material.

ARCHIVE & DOI UPDATE

Riemann Bandwidth-One Verification - permanent versioned record

Verified publication record (Zenodo screenshot, 9 October 2026)

Author	Prince Upadhyay
Zenodo version DOI	10.5281/zenodo.23244818
All-versions DOI	10.5281/zenodo.23244817
Resource type	Software; open-access files; custom scoped rights notice
Publication date	8 October 2026
GitHub source tag	v0.1.0 at 6d799bddd9a27ba79c4faf8c757f7985ceb6d4c2

Research claim boundaries carried into the deposit

- Exact finite identities and certificates apply only to their stated objects.
- The historical 68.820273% simple-critical-zero calculation is conditional.
- The 79.627% figure is a model frontier, not a proved zeta-zero result.
- The strict greater-than-79% theorem and the Riemann Hypothesis remain open.
- A DOI records an archive; it is not an originality or peer-review certificate.

Archive identity and citation notes

This publication addendum is part of the downloadable master dossier, not part of the frozen Zenodo v0.1.0 source ZIP. The original 18-page portfolio audit is preserved. In the public Zenodo interface, the metadata currently displays Version "v1" while the GitHub source release is tagged "v0.1.0". The creator is rendered "upadhyay, prince" with role "Other". These display differences do not change the archived source; metadata can be reviewed for consistency.

Public Zenodo record

Frozen GitHub release